Zhaoyang GUAN

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Education

Northwestern University — Evanston, IL, USA
MSc in Engineering Sciences & Applied Mathematics — 09/2025-01/2027 (Expected)
GPA: 3.62/4

University of Liverpool — Liverpool, UK
BSc in Applied Mathematics (First Class Honours) — 09/2023-07/2025

Xi'an Jiaotong-Liverpool University — Suzhou, China
BSc in Applied Mathematics — 09/2021-09/2023
GPA: 3.65/4

Research Interests

Machine Learning; Time-Series Analysis; Partial Differential Equations; Large Language Models (LLMs) and Natural Language Processing (NLP)

Publications

Guan, Z., Cao, H., Zhong, M., Ni, Y., Yang, E., Ai, L., Ding, K., & Shi, T. (2026). *Symphony-Coord: Emergent Coordination in Decentralized Agent Systems.* arXiv preprint arXiv:2602.00966. **TMLR (Under submission).**

Zhang, J., **Guan, Z**., Xia, Z., & Shi, N. (2026). *Delay-Guided Attention for Quality Prediction in Multistage Manufacturing.* **Journal of Quality Technology (Under submission; Co-First Author).**

Guan, Z., Chen, Y., & Ding, K. (2026). *TANGO: Advancing LLMs for Time-Series Anomaly Detection with Grounded Reasoning.* **ACL Rolling Review (ARR), August 2026 (Under Review).**

Yu, D., Li, Y., **Guan, Z.**, Lin, X., Luan, J., Rao, Z., Lan, X., Ran, Y., Lan, B., Zhai, N.-X., Du, H., Fu, J., Deng, W., Ni, Y., & Li, C. (2026). *SkillAudit: From Fixed-Suite Benchmarking to Skill-Centered Assessment.* **ACL Rolling Review (ARR), August 2026 (Under Review; Co-First Author).**

Zhu, Y., Xu, R., **Guan, Z.**, Zhang, K., Rudko, N., Zhang, C., Muenich, R., Candan, K. S., & Ding, K. (2026). *MUSTANG: Multi-Variable Spatio-Temporal Meta-Learning for Water Data Imputation.* **KDD 2026.**

Guan, Z., Li, J., Liu, X., & Shan, X. (2025). *Dynamic Adjacency Matrix Learning and Multi-Order Random Walk Aggregation for Alzheimer's Disease Diagnosis From Resting-State fMRI.* **IEEE Access, 13**, 119427–119442. https://doi.org/10.1109/ACCESS.2025.3587286

Guan, Z., Xie, J., Wu, S., & Liang, C. (2025). *Research on Predicting the Risk Level of Coal Mine Roof Accident Based on Machine Learning.* **Scientific Reports, 15**, 24028. https://doi.org/10.1038/s41598-025-07760-6

Luo, Z., **Guan, Z.**, & You, W. (2025). *GaitMAFF: Adaptive Multi-Modal Fusion of Skeleton Maps and Silhouettes for Robust Gait Recognition in Complex Scenarios.* **Computers, Materials & Continua.** https://doi.org/10.32604/cmc.2025.075704

Guan, Z., Li, J., & Liu, Y. (2024). *Flood Risk Research Based on BP Neural Network and K-Means Clustering Algorithm.* **IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE).** https://doi.org/10.1109/ICEACE63551.2024.10898258

Guan, Z. (2024). *Urban Temperature Prediction Model Based on CNN-Bi-LSTM Model.* **7th International Conference on Information Technologies and Electrical Engineering.** https://doi.org/10.1145/3717934.3717943

Research Experience

REAL Lab, Northwestern University — Prof. Kaize Ding

Graduate Research Assistant

2026 – Present

- Co-led research on large language models and vision-language models for time-series anomaly detection, focusing on both anomaly detection and precise temporal localization.
- Developed TANGO, a supervised multimodal framework that formulates time-series anomaly detection as a detect-and-ground task, jointly producing anomaly labels, localized intervals, structured evidence, and natural-language summaries.
- Designed a synthetic-to-real training pipeline that learns representative anomaly patterns from controlled synthetic data and transfers the learned grounding capability to real-world time series.
- Developed a self-calibrated optimization strategy that identifies false negatives, false positives, and boundary errors from model predictions and uses boundary-aware policy optimization to improve unresolved cases.
- Conducted extensive experiments across seven TSB-AD-U datasets and multiple Qwen-VL backbones, achieving a best mean F1 score of 63.75%, outperforming the strongest evaluated baseline by 24.28 percentage points.

Integrative AI Lab, Northwestern University — Prof. Naichen Shi

Graduate Research Assistant

2026 – Present

- Co-led research on time-series forecasting and quality prediction for multistage manufacturing systems, with a focus on time-varying delays between production stages.
- Developed Delay-Aware Multi-stage Forecasting (DAMF) to explicitly estimate sample-dependent inter-stage lag distributions and incorporate temporal delay information into final quality prediction.
- Designed an edge-level lag predictor with occurrence–magnitude factorization to distinguish whether a delay occurs and estimate its magnitude across adjacent manufacturing stages.
- Integrated predicted lag distributions into a differentiable DelayAlignment module and gated residual fusion mechanism to align and propagate information across successive production stages.
- Developed a segment- and duration-aware Viterbi decoding strategy for recovering stable and interpretable lag paths, and evaluated the framework on real-world LiquidSugar data and controlled-delay benchmarks.
- Achieved the best performance on the LiquidSugar quality-prediction task with RMSE 5.24, MAE 3.60, and (R^2=0.673), outperforming the strongest multistage forecasting baseline.

Internship Experience

Institute of Automation, Chinese Academy of Sciences (CASIA) 06/2025–09/2025

Research Assistant Beijing, China

- Contributed to the project "Whole-Brain Network Modeling for Neurological Disease Regulation," supporting research on brain network dynamics and neural regulation.
- Performed brain imaging data quality control and standardization; computed dynamic functional connectivity (dFC) from multiple datasets (structural and functional connectivity) to ensure reliable downstream modeling.
- Built whole-brain dynamical models based on the Dynamic Mean Field (DMF) framework by integrating structural connectivity with functional connectivity signals.
- Conducted parameter fitting and optimization for inter-regional coupling strength and neurodynamic features; ran simulations under varying noise levels and regulatory conditions to examine potential clinical relevance.
- Analyzed dynamical stability and interpretability metrics for DMF outputs; carried out systematic literature review on neural regulation methods and organized references for model selection and validation.